# Lab 7: Fuzzing for Input Validation Bugs

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## Approach

Each of the three functions in processor.py gets a Hypothesis test that feeds it arbitrary Unicode strings or None (st.text() | st.none()). Each call is wrapped in try/except: any exception fails the test and reports the input that caused it. Each test also checks that the result makes sense: an ISBN summary string, a depreciation factor between 0.1 and 1.0, or one of the five known condition multipliers (None means the input was rejected).

## Bugs found by fuzzing (screenshot 2)

| Function | Input | Exception |
|---|---|---|
| `verify_isbn10_format` | `':'` | `ValueError: invalid literal for int() with base 10: ':'` |
| `verify_isbn10_format` | `'0'` | `IndexError: string index out of range` |
| `verify_isbn10_format` | `''` | `IndexError: string index out of range` |
| `verify_isbn10_format` | `None` | `AttributeError: 'NoneType' object has no attribute 'replace'` |
| `calculate_depreciation_factor` | `'/'` | `ValueError: could not convert string to float: ''` |
| `calculate_depreciation_factor` | `''` | `ValueError: not enough values to unpack (expected 2, got 1)` |
| `calculate_depreciation_factor` | `None` | `AttributeError: 'NoneType' object has no attribute 'split'` |
| `lookup_condition_multiplier` | `''` | `KeyError: ''` |
| `lookup_condition_multiplier` | `None` | `AttributeError: 'NoneType' object has no attribute 'strip'` |

## Fixes in processor.py

- All three functions return None for anything other than a str, so None no longer crashes them.
- verify_isbn10_format: after removing hyphens and spaces, the input must be exactly 9 ASCII digits followed by a digit or X; otherwise it returns None. This handles '', '0' (too short) and ':' (not a digit).
- calculate_depreciation_factor: the input must split into exactly two parts on '/', both must convert to float, and both must be finite, with age >= 0 and lifespan > 0; otherwise it returns None. This handles '', '/', 'a/b', and also '3/0' (division by zero), negative ages and nan/inf, which are hardened too. The result always stays between 0.1 and 1.0.
- lookup_condition_multiplier: uses dict.get(), so an unknown grade like '' or 'FAIR' returns None instead of raising KeyError.
- main() prints a clear message (Invalid ISBN-10 format / Invalid age/lifespan / Unknown condition) instead of crashing. Valid inputs give exactly the same results as before.

**Result: all 3 Hypothesis tests now pass (screenshot 4). They also pass when re-run with 5,000 examples per test.**

## 1. Initial (buggy) processor.py execution

Run with the example inputs works, but the invalid input 'abc' crashes with ValueError.

```
PS C:\bookstore\Lab7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2): 0-306-40615-2
Group 306: Starts 0, Checksum 2

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): 3/10
Depreciation Factor: 0.7

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR): MINT
Multiplier: 1.25

PS C:\bookstore\Lab7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2): abc
Traceback (most recent call last):
  File "C:\bookstore\Lab7\processor.py", line 40, in <module>
    main()
  File "C:\bookstore\Lab7\processor.py", line 29, in main
    print(verify_isbn10_format(isbn_in))
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\bookstore\Lab7\processor.py", line 3, in verify_isbn10_format
    first_digit = int(cleaned[0])
                  ^^^^^^^^^^^^^^
ValueError: invalid literal for int() with base 10: 'a'
```

## 2. Failing Hypothesis test cases

python -m pytest test_processor.py against the buggy code: 3 failed, with Hypothesis reporting every distinct crashing input it found.

```
PS C:\bookstore\Lab7> python -m pytest test_processor.py
============================= test session starts =============================
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\bookstore\Lab7
plugins: anyio-4.14.2, hypothesis-6.170.14, platformdirs-4.13.0
collected 3 items

test_processor.py FFF                                                    [100%]

================================== FAILURES ===================================
_____________________ test_verify_isbn10_format_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\bookstore\Lab7\test_processor.py", line 5, in test_verify_isbn10_format_no_crash
  |     def test_verify_isbn10_format_no_crash(s):
  |                ^^^
  |   File "C:\Users\Asus\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2330, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 7, in test_verify_isbn10_format_no_crash
    |     result = verify_isbn10_format(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 3, in verify_isbn10_format
    |     first_digit = int(cleaned[0])
    |                   ^^^^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ':'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 9, in test_verify_isbn10_format_no_crash
    |     assert False, f"verify_isbn10_format crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: verify_isbn10_format crashed on ':' with: ValueError: invalid literal for int() with base 10: ':'
    | assert False
    | Failing test case: test_verify_isbn10_format_no_crash(
    |     s=':',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\bookstore\Lab7\test_processor.py:8
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 7, in test_verify_isbn10_format_no_crash
    |     result = verify_isbn10_format(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 5, in verify_isbn10_format
    |     check_char = cleaned[9].upper()
    |                  ~~~~~~~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 9, in test_verify_isbn10_format_no_crash
    |     assert False, f"verify_isbn10_format crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: verify_isbn10_format crashed on '0' with: IndexError: string index out of range
    | assert False
    | Failing test case: test_verify_isbn10_format_no_crash(
    |     s='0',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\bookstore\Lab7\test_processor.py:8
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 7, in test_verify_isbn10_format_no_crash
    |     result = verify_isbn10_format(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 3, in verify_isbn10_format
    |     first_digit = int(cleaned[0])
    |                       ~~~~~~~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
```

```
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 9, in test_verify_isbn10_format_no_crash
    |     assert False, f"verify_isbn10_format crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: verify_isbn10_format crashed on '' with: IndexError: string index out of range
    | assert False
    | Failing test case: test_verify_isbn10_format_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\bookstore\Lab7\test_processor.py:8
    +---------------- 4 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 7, in test_verify_isbn10_format_no_crash
    |     result = verify_isbn10_format(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 2, in verify_isbn10_format
    |     cleaned = isbn_str.replace('-', '').strip()
    |               ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'replace'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 9, in test_verify_isbn10_format_no_crash
    |     assert False, f"verify_isbn10_format crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: verify_isbn10_format crashed on None with: AttributeError: 'NoneType' object has no attribute 'replace'
    | assert False
    | Failing test case: test_verify_isbn10_format_no_crash(
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\bookstore\Lab7\test_processor.py:8
    +------------------------------------
_____________________ test_calculate_depreciation_factor_safe _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\bookstore\Lab7\test_processor.py", line 13, in test_calculate_depreciation_factor_safe
  |     def test_calculate_depreciation_factor_safe(s):
  |                    ^^^
  |   File "C:\Users\Asus\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2330, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 15, in test_calculate_depreciation_factor_safe
    |     result = calculate_depreciation_factor(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 10, in calculate_depreciation_factor
    |     age = float(age_str)
    |           ^^^^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 17, in test_calculate_depreciation_factor_safe
    |     assert False, f"calculate_depreciation_factor crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: calculate_depreciation_factor crashed on '/' with: ValueError: could not convert string to float: ''
    | assert False
    | Failing test case: test_calculate_depreciation_factor_safe(
    |     s='/',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 15, in test_calculate_depreciation_factor_safe
    |     result = calculate_depreciation_factor(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 9, in calculate_depreciation_factor
    |     age_str, life_str = book_age_str.split('/')
    |     ^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 17, in test_calculate_depreciation_factor_safe
    |     assert False, f"calculate_depreciation_factor crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: calculate_depreciation_factor crashed on '' with: ValueError: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_calculate_depreciation_factor_safe(
    |     s='',  # or any other generated value
    | )
```

```
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 15, in test_calculate_depreciation_factor_safe
    |     result = calculate_depreciation_factor(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 9, in calculate_depreciation_factor
    |     age_str, life_str = book_age_str.split('/')
    |                         ^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 17, in test_calculate_depreciation_factor_safe
    |     assert False, f"calculate_depreciation_factor crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: calculate_depreciation_factor crashed on None with: AttributeError: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_calculate_depreciation_factor_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_lookup_condition_multiplier_safe _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\bookstore\Lab7\test_processor.py", line 21, in test_lookup_condition_multiplier_safe
  |     def test_lookup_condition_multiplier_safe(s):
  |                  ^^^
  |   File "C:\Users\Asus\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2330, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 23, in test_lookup_condition_multiplier_safe
    |     result = lookup_condition_multiplier(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 24, in lookup_condition_multiplier
    |     return multipliers[grade]
    |            ~~~~~~~~~~~^^^^^^^
    | KeyError: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 25, in test_lookup_condition_multiplier_safe
    |     assert False, f"lookup_condition_multiplier crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: lookup_condition_multiplier crashed on '' with: KeyError: ''
    | assert False
    | Failing test case: test_lookup_condition_multiplier_safe(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\bookstore\Lab7\test_processor.py:24
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 23, in test_lookup_condition_multiplier_safe
    |     result = lookup_condition_multiplier(s)
    |              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\bookstore\Lab7\processor.py", line 16, in lookup_condition_multiplier
    |     grade = condition_code.strip().upper()
    |             ^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\bookstore\Lab7\test_processor.py", line 25, in test_lookup_condition_multiplier_safe
    |     assert False, f"lookup_condition_multiplier crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: lookup_condition_multiplier crashed on None with: AttributeError: 'NoneType' object has no attribute 'strip'
    | assert False
    | Failing test case: test_lookup_condition_multiplier_safe(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\bookstore\Lab7\test_processor.py:24
    +------------------------------------
========================= short test summary info ==========================
FAILED test_processor.py::test_verify_isbn10_format_no_crash - ExceptionGroup...
FAILED test_processor.py::test_calculate_depreciation_factor_safe - Exception...
FAILED test_processor.py::test_lookup_condition_multiplier_safe - ExceptionGr...
============================ 3 failed in 2.44s =============================
```

## 3. Fixed processor.py execution

Valid inputs give the same results as before; invalid and edge-case inputs are reported instead of crashing.

```
PS C:\bookstore\Lab7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2): 0-306-40615-2
Group 306: Starts 0, Checksum 2

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): 3/10
Depreciation Factor: 0.7

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR): MINT
Multiplier: 1.25

PS C:\bookstore\Lab7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2): abc
Invalid ISBN-10 format

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): 3/0
Depreciation Factor: Invalid age/lifespan

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR): FAIR
Multiplier: Unknown condition

PS C:\bookstore\Lab7> python processor.py
==== Running verify_isbn10_format ====
Enter ISBN-10 (e.g. 0-306-40615-2):
Invalid ISBN-10 format

==== Running calculate_depreciation_factor ====
Enter age/lifespan in years (e.g. 3/10): -2/10
Depreciation Factor: Invalid age/lifespan

==== Running lookup_condition_multiplier ====
Enter condition (e.g. MINT, GOOD, POOR):   good
Multiplier: 0.6
```

## 4. Successful Hypothesis test execution

python -m pytest test_processor.py against the fixed code: 3 passed.

```
PS C:\bookstore\Lab7> python -m pytest test_processor.py
============================= test session starts =============================
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\bookstore\Lab7
plugins: anyio-4.14.2, hypothesis-6.170.14, platformdirs-4.13.0
collected 3 items

test_processor.py ...                                                    [100%]

============================== 3 passed in 1.83s ==============================
```